

# Foundations of Reinforcement Learning: MDPs and Q-Learning

## Reinforcement Learning Primer

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## 1 The Markov Decision Process (MDP) Framework

The control problem in reinforcement learning is formally defined using a Markov Decision Process, represented by the tuple  $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ .

### 1.1 Components

- **State Space ( $\mathcal{S}$ ):** The set of all possible environment configurations.
- **Action Space ( $\mathcal{A}$ ):** The set of all available moves for the agent.
- **Transition Probability ( $P$ ):** The mapping  $P(s'|s, a)$ , defining the probability of reaching state  $s'$  from state  $s$  via action  $a$ .
- **Reward Function ( $R$ ):** The immediate feedback  $R(s, a)$  received after taking an action.
- **Discount Factor ( $\gamma$ ):** A constant  $\gamma \in [0, 1)$  that balances immediate and future rewards.

### 1.2 The Objective Function

The agent's goal is to maximize the expected discounted return  $G_t$ :

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (1)$$

## 2 The Q-Learning Algorithm

Q-learning is a model-free, off-policy algorithm used to find the optimal action-value function  $Q^*(s, a)$ .

### 2.1 The Bellman Optimality Equation

The optimal Q-value, denoted by  $Q^*(s, a)$ , is the maximum expected return achievable from a given state-action pair  $(s, a)$ , and it satisfies the Bellman optimality equation:

$$Q^*(s, a) = \mathbb{E} \left[ r + \gamma \max_{a'} Q^*(s', a') \mid s, a \right], \forall (s, a) \in \mathcal{S} \times \mathcal{A} \quad (2)$$

where  $s'$  is the next state resulting from taking action  $a$  in state  $s$ . We denote

$$\left[ r + \gamma \max_{a'} Q^*(s', a') \mid s, a \right] = F(Q^*, s, a, \omega), \forall (s, a) \in \mathcal{S} \times \mathcal{A} \quad (3)$$

where  $\omega$  represents the randomness in the environment. Then, the optimal Q-value can be expressed as fixed point of the stochastic Bellman operator  $F[Q]$ :

$$Q^*(s, a) = \mathbb{E} [F(Q^*, s, a, \omega)], \forall (s, a) \in \mathcal{S} \times \mathcal{A}. \quad (4)$$

## 2.2 Temporal Difference (TD) Update Rule

It is well-known in the literature of Stochastic Approximation that the fixed point of  $x = \mathbb{E}[f(x, \omega)]$  can be found by the following iterative procedure: For an appropriate condition of  $f$  and a sequence of i.i.d. samples  $\{\omega_n\}$ , the sequence  $\{x_n\}$  defined by

$$x_{n+1} = (1 - \alpha)x_n + \alpha f(x_n, \omega_n) \quad (5)$$

where  $\alpha$  is the learning rate. Applying this result to the Bellman optimality equation, we can derive the Q-learning update rule.

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (6)$$

where  $\alpha$  is the **learning rate**.

## 3 Control and Policy

Once the Q-values converge to  $Q^*$ , the optimal policy  $\pi^*$  is derived by selecting the action with the highest value in each state:

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a) \quad (7)$$

The above Q-learning algorithm is pure exploitation or zero-greedy. In practice, we often use an  $\epsilon$ -greedy policy to balance exploration and exploitation during learning. The  $\epsilon$ -greedy policy selects a random action with probability  $\epsilon$  and the action with the highest Q-value with probability  $1 - \epsilon$ .